

General Information

Chemical Designation:

PPSU (Polyphenylsulfone)

Fillers:

Unfilled

Color:

Tan or Black

Specific Gravity:

1.29

PPSU is an amorphous high performance thermoplastic offering better impact resistance and chemical resistance than polysulfone and polyetherimide (Ultem® PEI). Its natural color is (bone) white but is available in colors on a custom run basis. PPSU offers superior hydrolysis resistance when compared to other amorphous thermoplastics as measured by steam autoclaving cycles to failure. PPSU has virtually unlimited steam sterilizability. This makes it an excellent choice for medical devices where steam autoclaves used for sterilization. It also resists common acids and bases over a broad temperature range.

PPSU is commonly used in sterilization trays, dental and surgical instrument handles and in fluid handling coupling and fitting applications. It is suitable for use in electronic assembly equipment and devices that must withstand soldering temperatures.

Technical Information

Specification	Test	Value	Units
Specific Gravity, 73°F	D792	1.29	–
Tensile Strength @ Yield, 73°F	D638	11,500	psi
Tensile Modulus of Elasticity, 73°F	D638	340,000	psi
Tensile Elongation (at break), 73°F	D638	60	%
Flexural Strength, 73°F	D790	13,500	psi
Flexural Modulus of Elasticity	D790	350,000	psi
Shear Strength, 73°F	D732	9,000	psi
Compressive Strength – Ultimate			psi
Compressive Strength at 2% Deformation	D695	2000	psi
Compressive Strength at 10% Deformation	D695	14,350	psi
Deformation Under Load			%
Compressive Modulus of Elasticity, 73°F	D695	234,000	
Compressive Strength to Laminate (Modulus)			psi
Compressive Strength to Laminate (Yield)			psi
Compressive Strength to Laminate (Ultimate)			psi
Hardness, Durometer (Shore "D" scale)	D2240		
Hardness, Rockwell (Scale as noted)	D785	M96 (R120)	Rockwell M/R
Izod Impact, Notched @ 73°F	D256 Type A	3.0	ft.lbs/in. of notch
Coefficient of Friction (Dry vs Steel) Static	PTM55007		
Coefficient of Friction (Dry vs Steel) Dynamic	PTM55007	–	
Maximum Static Bearing Load (P)	PTM55007		psi
Maximum Unlubricated No Load Bearing Velocity (V)	PTM55007		ft/minute
Maximum Limiting PV (Unlubricated)	PTM55007	–	psi x ft/min.
Wear Factor "K" x 10-10	PTM55010	>1,000	Cubic in.-min/ft.lbs.hr
Sand Wheel Wear/Abrasion Test			UHMW=100
Minimum Mating Surface Hardness			Rockwell (Brinnell)
Coefficient of Linear Thermal Expansion	E-831(TMA)	3.10	in/in/°F x 10-5
Coefficient of Thermal Expansion // to Laminates	E-831(TMA)	3.10	in/in/°F x 10-5
Coefficient of Thermal Expansion I to Laminates	E-831(TMA)	3.10	in/in/°F x 10-5
Softening Point			°F
Heat Deflection Temperature 264 psi	D648	405	°F
Embrittlement Temperature			°F Min.
Continuous Service Temperature in Air		300	°F Max.
Short Term Service Temperature		360	°F Max.
Tg-Glass Transition (Amorphous)	D3418	428	°F
Melting Point (Crystalline) Peak	D3418	N/A	°F
Thermal Conductivity	F433	1.74	BTU-in/(hr/ft2°F)
Dielectric Strength Short Term	D149	360	Volts/mil
Surface Resistivity	D257		ohm/cm
Volume Resistivity	D257	9x10^15	ohm/cm
Dielectric Constant, 60 Hz	D150	3.44	
Dissipation Factor, 106 Hz	D150	0.0017	
Flammability @ 3.1mm(1/8 in.) UL94	UL94	V-O	
Arc Resistance			seconds
Water Absorption, Immersion 24 Hours	D570 (2)	.37	%
Water Absorption, Immersion Saturation	D570 (2)	1.10	%
Machinability Rating		3	1=easy, 10=difficult
Rod Diameter Availability (Off the Shelf)		.50-3.0	inches
Sheet Thickness Availability (Off the Shelf)		.25-2.5	inches
Characteristics / Attributes	High Resistance to Steam Autoclaving / Impact Resistant		

Thank you for your interest in our materials. All statements, technical information and recommendations presented are in good faith, based upon tests believed to be reliable and practical field experience. Poly-Tech is not responsible for its accuracy or completeness. It is our recommendation and the customer's responsibility to determine the suitability of any material for any given application.